

# Same Area, Different Borders



**Grade 4 • Mathematics**  
**NC.4.MD.3**

Distinguish area from perimeter.



# What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



## Watch the reasoning

**A rectangular reading space is 12 m long and 10 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?**

Area =  $12 \times 10 = 120 \text{ m}^2$  for covering. Perimeter =  $2(12+10) = 44 \text{ m}$  for the border.



## Build a second model together

A rectangular reading space is 7 m long and 4 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Discuss: What should the first step find?**



## Your turn • 1

A rectangular reading space is 8 m long and 8 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Represent → solve → explain → check**



## Your turn • 2

A rectangular reading space is 8 m long and 10 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Represent → solve → explain → check**



## Your turn • 3

A rectangular reading space is 8 m long and 4 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Represent → solve → explain → check**



## Your turn • 4

A rectangular reading space is 9 m long and 10 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Represent → solve → explain → check**



## Exit question • Explain independently

A rectangular reading space is 9 m long and 8 m wide. Find its area and perimeter. Which answer measures floor covering, and which measures a border?

**Include a check. What would you teach someone else?**



# Teacher review and next steps

**1. 64 m<sup>2</sup>; 32 m**

**2. 80 m<sup>2</sup>; 36 m**

**3. 32 m<sup>2</sup>; 24 m**

**4. 90 m<sup>2</sup>; 38 m**

Exit: 72 m<sup>2</sup>; 34 m

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

